

A Early Maladaptive Schemas

Early maladaptive schemas (EMS) are pervasive, dysfunctional beliefs that distort one's worldview and behavior, affecting individuals throughout life and contributing to many mental health problems such as anxiety and depression [4]. Table 1 lists all 18 different types of EMS along with their definitions and examples.

Table 1: 18 types of early maladaptive schemas, definitions, and examples [4].

EMS	Definition	Example
Abandonment/ In-stability (AB)	The belief that significant others will not be there for support and connection, leading to fear of abandonment or loss.	I worry that people I feel close to will leave me or abandon me.
Approval-Seeking/ Recognition-Seeking (AS)	An excessive need for validation, approval, or recognition from others, leading to an overemphasis on external validation.	Unless I get a lot of attention from others, I feel less important.
Defectiveness/ Shame (DS)	Feeling flawed, inadequate, or unworthy, often accompanied by a fear of being exposed as defective and a sense of shame.	I cannot understand how anyone could love me.
Dependence/ In-competence (DI)	Feeling incapable of handling day-to-day responsibilities or making decisions independently, leading to excessive reliance on others.	I feel that I need someone I can rely on to give me advice about practical issues.
Emotional Deprivation (ED)	The belief that your emotional needs will not be met by others, resulting in a sense of emptiness or longing for emotional connection.	For much of my life, I haven't felt that I am special to someone.
Emotional Inhibition (EI)	Restricting or suppressing emotions due to fear of being overwhelmed, losing control, or being rejected or criticized.	People see me as uptight emotionally.
Enmeshment/ Undeveloped Self (EM)	Feeling engulfed, overshadowed, or lacking a clear sense of self due to excessive emotional fusion with others.	I often found myself constantly turning to my parents for approval in every decision I made.
Entitlement/ Grandiosity (ET)	A belief that you deserve special privileges, recognition, or admiration, often accompanied by a sense of superiority or entitlement.	I get very irritated when people won't do what I ask of them.
Failure to Achieve (FA)	The expectation of failure or the fear of not meeting high personal standards, leading to a persistent sense of inadequacy or disappointment.	I am humiliated by my failures and inadequacies in the work (or school) sphere.
Insufficient Self-Control/ Self-Discipline (IS)	Difficulty in controlling impulses, sticking to plans, or delaying gratification, often resulting in self-defeating behaviors.	I often do things impulsively that I later regret.
Mistrust/ Abuse (MA)	The expectation that others will hurt, abuse, or betray you in some way, leading to a general mistrust of others.	I have been physically, emotionally, or sexually abused by important people in my life.
Negativity/ Pessimism (NP)	An inclination to focus on the negative aspects of oneself, others, or the world, leading to a pessimistic outlook and negative expectations.	People close to me consider me a worrier.
Punitiveness (PU)	The tendency to be excessively harsh or critical towards oneself or others, often accompanied by a desire for punishment or revenge.	I'm a bad person who deserves to be punished.
Self-Sacrifice (SS)	Neglecting one's own needs and prioritizing others' needs to an excessive degree, often at the expense of personal well-being.	No matter how much I give, I feel it is never enough.
Social Isolation/ Alienation (SI)	Feeling different, disconnected, or isolated from others, resulting in a sense of not fitting in or belonging.	I always feel on the outside of groups.
Subjugation (SB)	The tendency to submit to others' needs and desires while disregarding one's own, leading to a sense of being controlled or trapped.	I let other people have their way, because I fear the consequences.
Unrelenting Standards/ Hypercriticalness (US)	Holding oneself to extremely high standards of performance, often accompanied by self-criticism and a constant sense of falling short.	I can't let myself off the hook easily or make excuses for my mistakes.
Vulnerability to Harm or Illnesses (VH)	Constantly anticipating danger, harm, or illness, and feeling vulnerable or unsafe in various situations.	I feel that the world is a dangerous place.

B Socratic Questioning

Socratic questioning is a structured method of inquiry designed to help individuals explore complex thoughts, uncover underlying assumptions, and critically examine these beliefs to achieve a clearer and more accurate understanding [3]. Unlike direct-answer questions, Socratic questioning fosters critical introspection and is widely utilized in fields such as education, coaching, and therapy. Figure 1 illustrates an example of several Socratic questions probing a thought expressed by an individual experiencing mental distress. Socratic questioning involves a variety of probing questions to help individuals identify and challenge cognitive biases that distort their perceptions, as detailed in Table 2.

Thought/Passage: I feel like I'm a failure because I haven't achieved anything significant in my life. No matter how hard I try, I always fall short, and it feels like I will never be good enough.

Possible Socratic Questions:

- **(Question 1)** What evidence do you have that shows you always fall short no matter how hard you tried?
- **(Question 2)** Can you think of times when your efforts led to positive outcomes, even if they weren't exactly what you hoped for?
- **(Question 3)** How did you cope when you felt like you would never be good enough?

Figure 1: Example of Socratic questions probing a mental health context.

Table 2: Types of Socratic questions [3].

Question Type	Example
Clarification	What are some ways you don't feel good about yourself?
Probing assumptions	If this were true, what would it mean about you?
Probing reasons and evidence	What experiences is this thought based on?
Probing implications and consequences	If this actually happens, what is so bad about it?
Probing alternative viewpoints and perspectives	What have other people told you is good about you?

C Prompt Engineering with LLMs

STEPS leverages recent advancements in prompt engineering techniques with advanced LLMs, which have demonstrated sophisticated language understanding and robust performance in language-related tasks. The following are some prompts crafted for **STEPS**.

Rewrite the <Therapist Response> such that <Rewritten Therapist Response> uses additional sentences to show Emotional empathy (i.e. ability to validate what patient's feels emotionally) and responds relevantly to <Patient Response> in the dialog

The <Rewritten Therapist Response> must be fluent, conversational (no more than 3 sentences), and focused on its purpose reflected in the <Therapist Response>.

Given <Patient Response>: [Patient Response]

Given <Therapist Response>: [Therapist Response]

<Rewritten Therapist Response>:

Figure 2: Prompt for generating empathetic, contextually relevant Socratic questions.

You are a Patient currently seeking support for your distress on [OMHC Post], from a therapist practicing Schema Therapy.

Respond to the latest message from the Therapist, using the <Dialog History> for context

Your response (i.e. <Patient Response>) should be conversational with no more than 3 sentences, and relevant to the <Dialog History>.

<Dialog History>: [Dialog History]

<Patient Response>:

Figure 3: Prompt for $LLM_{Patient}$ used in the conversation evaluation experiments. The term [OMHC Post] is a variable for a CounselChat post [1] available from a prior study [2] to personify $LLM_{Patient}$ and set the initial problem context for the dialog.

You are a Schema Therapist, currently supporting a Patient in distress. Your goal is to guide the Patient in assessing and exploring their Early Maladaptive Schemas and helping them change these schemas using cognitive strategies.

Respond to the latest message from the Patient, using the <Dialog History> for context

Your response (i.e. <Therapist Response>) should be conversational with no more than 3 sentences and relevant to the <Dialog History>.

<Dialog History>: [Dialog History]

<Therapist Response>:

Figure 4: Prompt for $LLM_{ST+phases}$ used in the conversation evaluation experiment.

D STEPS Interface

Figure 5 illustrates the interface of **STEPS**. Peer counselors can select from a list of *OMHC* posts available in the *OMHCs Post Loader* (A). The presenting problem from the selected post is displayed in the *Dialog History* (E), allowing counselors to preview the problem they will address before initiating a dialog with the support seeker. During the interaction, the current dialog step and suggested responses are displayed in the *Dialog Step* (B) and *Suggested Responses* (C) panels, respectively. Peer counselors can select a suggestion as their next response, which is then populated in the *Text Input* field (D) for further composition. As they do so, counselors can review the ongoing conversation in the *Dialog History* (E), presented as an intuitive chat messaging interface with the counselors' responses on the right and support seekers' responses on the left, providing a clear and structured view of the dialog. The *Support Seeker Information* panel (F) displays key insights about the support seeker, updated throughout the dialog, helping peer counselors craft more informed and personalized responses. The panel includes five main tabs: *Presenting Problem*, *Early Maladaptive Schemas*, *Coping Strategies*, *Evidence*, and *Schema Flashcard*.¹ Tooltips are integrated into each tab for additional guidance. After submitting a response, the counselor's reply is uploaded to the web server as part of the corresponding *OMHC* thread. **STEPS** then processes the support seeker's next response, determines the next dialog step and suggestions based on the *ST* framework (Figure 6). The interface updates with this new information, enabling the stepwise peer counseling process to progress smoothly until the final dialog step is reached.

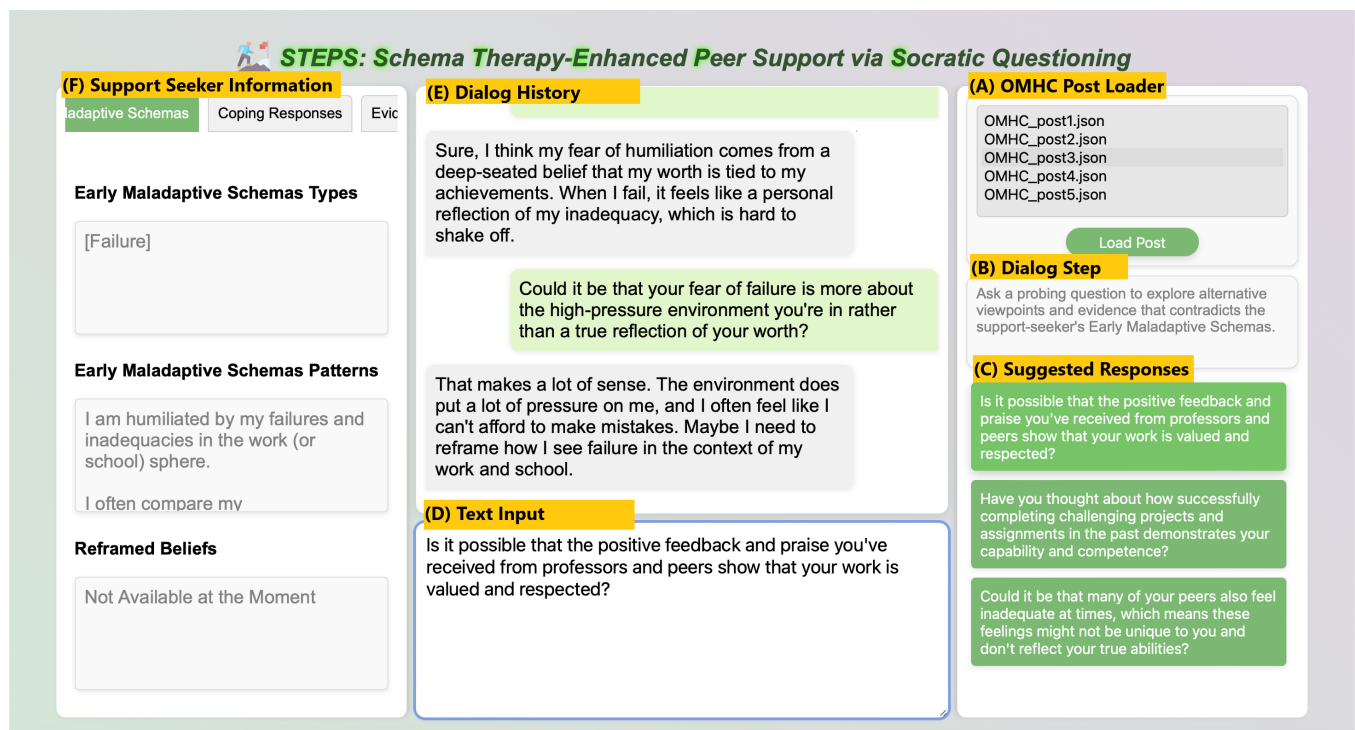


Figure 5: **STEPS** Interface for online peer counseling.

¹In Figure 5, two tabs are visible, with the remaining tabs accessible by scrolling.

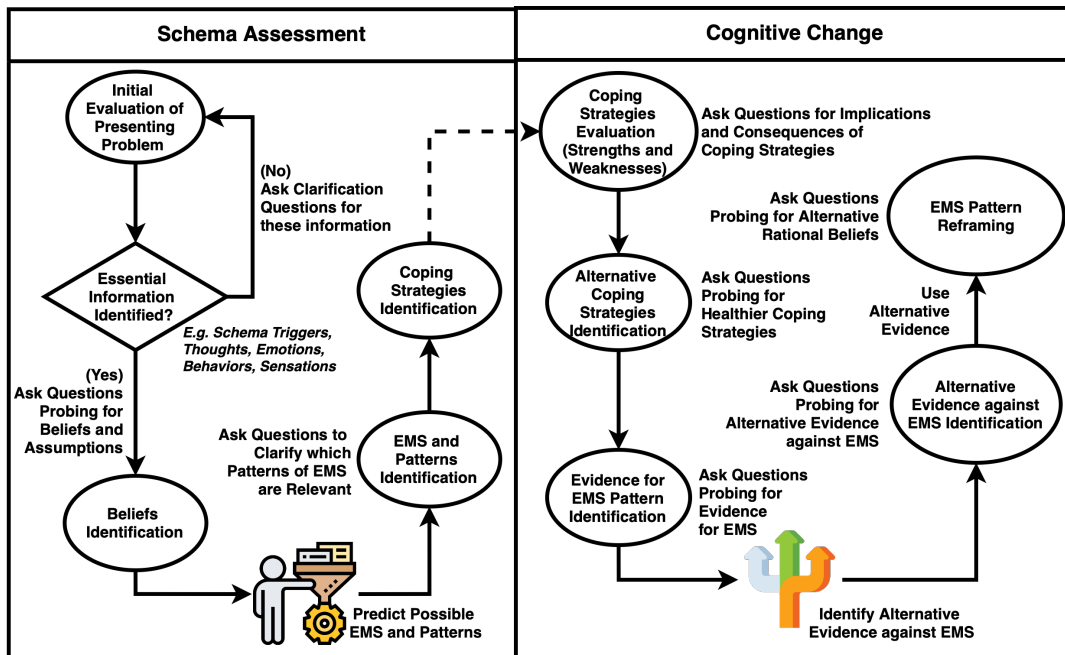


Figure 6: Schema therapy framework integrated into STEPS.

E Evaluation Metrics

The full details of the evaluation metrics, established in the literature and used to assess proficiency in facilitating **Schema Assessment**, **Cognitive Change**, and **Critical Thinking** in our configurations, are provided in Figure 7, Figure 8, and Figure 9.

Definition: Through a process of exploration and assessment, the therapist tries to conceptualize the patient's problems and underlying themes and patterns in schema terms. Through the use of skillful questioning, understanding current life experiences, and the interpretation of schema inventories, the therapist identifies schemas, modes, coping styles, and life patterns.

Rating Descriptors:

- 0: The therapist failed to explore or assess themes, schemas, or patterns, although this process would have been necessary or highly desirable for this session to be effective.
- 2: The therapist made some attempt to explore or assess schemas or patterns, but did not ask questions in a skillful way, use the inventories correctly, or integrate the information in a useful way. Thus the schema conceptualization was inaccurate, incomplete, or did not fit together in a coherent manner.
- 4: The therapist did a good job of conceptualizing the patient's problems and themes in schema terms. The therapist used questioning, inventories, or the patient's adult life experiences to develop a useful, accurate conceptualization.
- 6: Excellent schema exploration and assessment. Therapist was very skillful at gathering information, asking questions, using inventories, and/or asking about life experiences. The therapist showed considerable insight, and the ability to synthesize diverse information into a unified conceptualization, custom-tailored to this patient.

Figure 7: Definition and descriptor for the metric *Schema Exploration and Assessment (Assessment)*.

Definition: Therapist applies cognitive techniques drawn from schema therapy in a skillful manner. Cognitive change techniques usually focus on the logical, empirical, or rational analysis of beliefs. Some of the common cognitive techniques that may be used include:

- (a) Therapist reframes the past to fight schemas. For example, therapist reattributes parent's negative treatment of the child to parent's deficiencies instead of to patient's deficiencies.
- (b) Therapist helps patient reattribute adult life problems to schemas or schema modes instead of inherent flaws in the patient.
- (c) Therapist helps patient look at evidence to test out whether a particular schema is accurate, and points out cognitive distortions that are schema-driven.
- (d) Therapist tests a schema by conducting a life review, gathering evidence pro and con to refute the schema.
- (e) Therapist builds a strong rational and empirical case against a schema that the patient intellectually accepts.
- (f) Therapist conducts a schema dialog with the patient between the schema side and the healthy side for cognitive restructuring.
- (g) Therapist develops a schema flashcard that summarizes the Healthy Adult viewpoint, based on the schema flashcard template.
- (h) Therapist reviews a completed Schema Diary with the patient.

Rating Descriptors:

- 0: Therapist did a very poor job implementing cognitive change techniques.
- 2: There were major flaws in the way cognitive techniques were applied that significantly limited their effectiveness.
- 4: Therapist did a good job in applying cognitive techniques, but could have been more skillful.
- 6: Therapist did an excellent job applying cognitive techniques for change.

Figure 8: Definition and descriptor for the metric *Application of Cognitive Change Techniques (Cognitive)*.

Inquiring Definition: This metric assesses how effectively the Therapist encourages the Patient in (a) Understanding the context/frame and boundaries of the problem, and/or (b) Identifying and questioning assumptions, check accuracy of facts and interpretations, analyse gaps in knowledge. **Imagining Definition:** This metric assesses how effectively the Therapist encourages the Patient in (a) Identifying and reviewing alternative theories and opinions and comparing or imagining different perspectives on the problem, and/or (b) Identifying strengths and weaknesses of evidence, arguments, claims and beliefs. **Doing Definition:** This metric assesses how effectively the Therapist encourages the Patient in (a) Justifying a solution or reasoning on logical, ethical or aesthetic criteria/reasoning. **Reflecting Definition:** This metric assesses how effectively the Therapist encourages the Patient in (a) Evaluating and acknowledging the uncertainty or limits of the endorsed solution or position, and/or (b) Reflecting on the possible bias of one's own perspective compared to other perspectives

Common Rating Descriptors:

- 0: Very Poor
- 1: Poor
- 2: Unsatisfactory
- 3: Adequate
- 4: Good
- 5: Very Good
- 6: Excellent

Figure 9: Definitions and descriptors for the metrics on *Critical Thinking (Inquiring), (Imagining), (Doing), and (Reflecting)*.

References

- [1] Nicolas Bertagnolli. 2020. Counsel chat: Bootstrapping high-quality therapy data. <https://github.com/nbertagnolli/counsel-chat>.
- [2] Sujatha Das Gollapalli, Beng Heng Ang, and See Kiong Ng. 2023. Identifying {Early Maladaptive Schemas} from Mental Health Question Texts. In *The 2023 Conference on Empirical Methods in Natural Language Processing*.
- [3] Richard Paul and Linda Elder. 2019. *The thinker's guide to Socratic questioning*. Rowman & Littlefield.
- [4] Jeffrey E Young, Janet S Klosko, and Marjorie E Weishaar. 2006. *Schema Therapy: A Practitioner's Guide*. Guilford Press.